

Muzammal Hussain

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• ABOUT ME

Full-Stack Software Engineer | Agentic AI Engineer with 3+ years of professional experience building scalable SaaS platforms, cloud-native applications, and intelligent software solutions. Experienced in designing and developing modern web applications using TypeScript, Node.js, NestJS, React.js, PostgreSQL, and Python, with expertise across the full software development lifecycle.

Specialized in integrating Agentic AI and LLM-powered capabilities into production applications using OpenAI APIs, OpenAI Agents SDK, LangChain, LangGraph, CrewAI, and Model Context Protocol (MCP). Skilled in developing multi-agent systems, Retrieval-Augmented Generation (RAG) pipelines, AI workflow automation, and tool-calling architectures that enable intelligent, context-aware user experiences.

Proven experience delivering secure, scalable backend systems, multi-tenant SaaS platforms, GIS-based applications, cloud deployments on AWS and Google Cloud Platform, and production-ready APIs. Passionate about building reliable software that combines modern engineering practices with practical AI to solve complex real-world problems.

• WORK EXPERIENCE

31/10/2024 - CURRENT - E11, ISLAMABAD, PAKISTAN

SOFTWARE ENGINEER FUTURE NOSTICS

- Design and develop scalable backend services and modern frontend features for AEDI, a multi-tenant municipal planning SaaS platform used by Canadian municipalities for zoning and land development assessments.
- Build HIPAA-compliant healthcare solutions, including dynamic form builders, workflow automation, secure case management platforms, and enterprise-grade backend services for healthcare organizations.
- Engineer cloud-native applications using NestJS, React.js, FastAPI, PostgreSQL, Redis, Docker, AWS ECS, and Google Cloud Platform (GCP), ensuring scalability, reliability, and high availability.
- Design and implement secure authentication, authorization, and Role-Based Access Control (RBAC) across enterprise SaaS applications.
- Integrate third-party services including Stripe, GIS platforms, payment providers, and external APIs to extend platform capabilities and streamline business workflows.
- Collaborate with product managers, designers, and engineering teams in Agile environments, contributing to architecture decisions, code reviews, and the delivery of production-ready software.
- Developed the Sales Management System, a SaaS platform for managing products,

inventory, customers, orders, and sales operations by designing scalable backend APIs and business workflow automation.

- Developed Sidekick, an Agentic AI assistant using Python, LangChain, and LangGraph, capable of planning multi-step tasks, invoking external tools, supporting human-in-the-loop interactions, and evaluating responses through graph-based workflows.
- Built Autonomous Trading Floor, a multi-agent AI platform using Python, OpenAI Agents SDK, and Model Context Protocol (MCP), enabling specialized AI agents to collaborate on financial research, market analysis, and custom MCP server integration through autonomous agent orchestration.

31/05/2023 - 30/11/2024 - MODEL TOWN, LAHORE, PAKISTAN

BACKEND DEVELOPERARGONTEQ

- Developed scalable backend services and RESTful APIs using Node.js, NestJS, Express.js, MongoDB, and PostgreSQL for SaaS, fintech, healthcare, gaming, and education platforms.
- Built Willo Box, an AI-powered document intelligence platform using OpenAI API, LangChain, vector embeddings, and semantic search, enabling users to retrieve context-aware insights from uploaded documents.
- Designed secure authentication, authorization, and Role-Based Access Control (RBAC) systems for enterprise applications.
- Contributed to backend development of a cryptocurrency exchange platform, implementing user management, transaction processing, and secure API integrations.
- Developed backend services for pharmacy marketplace, restaurant engagement, gaming, and educational CRM platforms, delivering scalable business solutions across multiple industries.
- Integrated third-party services including Google Reviews, payment gateways, and external APIs to enhance application functionality.
- Worked closely with frontend developers, QA engineers, and product managers to deliver high-quality software through Agile development practices.

• **SKILLS**

Frontend

React.js | TypeScript | Vite | Bootstrap | Tailwind CSS | Zustand | TanStack Router/Query | Shadcn/UI

Backend

JavaScript | Express.js | Node.js | NestJS | GraphQL | Python | FastAPI | RabbitMQ | WebSockets | Microservices Architecture

Databases

MongoDB | PostgreSQL | MySQL | Redis | Vectore Database | Firestore

Version Control & Collaboration

Git | GitHub | GitLab | GitHub Actions | Husky | Jira | ClickUp | Slack

Cloud & DevOps

Docker | AWS (EC2, ECS, S3) | Google Cloud Platform (Cloud Run, Artifact Registry, GCS)

Testing

Jest | ESLint | Prettier | Storybook | Unit Testing | Integration Testing | Postman Automation Testing | Playwright

AI & Automation

OpenAI | LangChain | LangGraph | CrewAI | OpenAI Agents SDK | Model Context Protocol (MCP) | Agentic AI | Multi-Agent Systems | AI Agent Orchestration | Retrieval-Augmented Generation (RAG)

● **PROJECTS**

AEDI — Municipal Planning & Zoning SaaS

Technologies: React.js, NestJS, FastAPI, PostgreSQL, ArcGIS, OpenAI, AWS ECS, Docker

- Developed a multi-tenant SaaS platform for Canadian municipalities to automate zoning and land development assessments. Built scalable backend services, AI-powered regulatory analysis workflows, GIS integrations, and cloud-native infrastructure using OpenAI, RAG, and ArcGIS.

Case Management Hub (CMH)

Technologies: NestJS, PostgreSQL, Google Cloud Platform

- Developed secure backend services for a HIPAA-compliant case management platform, implementing configurable dynamic forms, role-based access control, workflow automation, and scalable APIs for healthcare organizations.

Sales Management System

Technologies: React.js, NestJS, PostgreSQL, TypeScript

- Developed a SaaS-based sales and inventory management platform for e-commerce businesses, enabling product management, inventory tracking, order processing, and sales operations. Designed secure backend APIs, implemented authentication and Role-Based Access Control (RBAC), and built scalable workflows to support day-to-day business operations.

Willo Box – AI Document Intelligence Platform

Technologies: Node.js, OpenAI API, LangChain, Vector Database

- Developed an LLM-powered document intelligence platform that enables users to upload PDF documents and retrieve accurate, context-aware answers using Retrieval-Augmented Generation (RAG), semantic search, and vector embeddings.

Monday.com Automation Platform

Technologies: Node.js, TypeScript, NestJS, Monday.com API, Webhooks

- Developed backend services and custom integrations for the Monday.com platform, automating business workflows through event-driven webhooks, custom applications, and third-party API integrations. Built scalable services to synchronize data, automate repetitive processes, and improve operational efficiency across customer workflows.

Obon – Easy Tipping Platform

Technologies: Node.js, NestJS, React.js, PostgreSQL, Stripe

- Developed a digital tipping platform that enables businesses to accept cashless tips through secure online payments. Built scalable backend APIs, integrated Stripe for payment processing, implemented authentication and role-based access control (RBAC), and developed responsive frontend features to provide a seamless tipping experience for customers and businesses.

Autonomous Trading Floor – Multi-Agent AI Platform

Technologies: Python, OpenAI Agents SDK, MCP (Model Context Protocol)

- Built a multi-agent trading platform where specialized AI agents collaborated to perform market research, analyze financial data, and access custom MCP servers for account management and market intelligence using autonomous agent orchestration.

Sidekick – Agentic AI Assistant

Technologies: LangChain, LangGraph, Python

- Developed an intelligent AI assistant capable of planning tasks, invoking external tools, evaluating responses against success criteria, and supporting human-in-the-loop interactions through graph-based agent workflows.

● **EDUCATION & TRAINING**

07/10/2021 - 31/05/2025 - LAHORE

BS SOFTWARE ENGINEERING- VIRTUAL UNIVERSITY OF PAKISTAN

Final grade: 3.69/4

22/05/2014 - 05/04/2016

F.SC PRE ENGINEERING- GOVT. DEGREE COLLEGE

Final grade: 82%